

Yile Technology: Bringing innovative, highly available online casino games to players worldwide

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[About the Technology](#)



About Yile Technology

Yile Technology is a Taiwanese mobile game developer dedicated to innovating online casino games. As one of the major online casino game providers in Taiwan, it has acquired more than 1 million members on Online808, its flagship game, and plans to enter the global market by releasing a new game by the end of 2021 and another Taiwan-specific game by early 2022.

Industries: Gaming, Media & Entertainment

To provide game
the world with the
gaming experience
Technology lever
Google Cloud to
games at ease, e

service stability, &
app response time

Location: Taiwan

(<https://cloud.google.com/kubernetes-engine/>),

Google Cloud results

- Supports 1 million gamers and counting with highly scalable Pub/Sub
- Ensures service stability during traffic surges with the autoscaling feature of Google Kubernetes Engine
- Enables new game development and frequent updates with zero downtime on

Shorter
response
by half



About CloudMile

CloudMile is a leading AI and cloud technology company in Asia that empowers businesses to accelerate digital transformation and drive growth. Leveraging machine learning and big data analysis, CloudMile has assisted more than 400 clients with business forecasts and industrial upgrades. A Google Cloud Premier Partner, the company has become a Google Cloud Managed Service Provider (MSP) covering Taiwan, Hong Kong, and Singapore and was the Public Sector Partner of the Year for APAC Region in 2020.

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Google Cloud (<https://cloud.google.com/>)

Anthos (<https://cloud.google.com/anthos/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)


Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Google
Kubernetes
Engine
• Frees up
more IT
workforce
for
development
and
innovation
with fully
managed
and
automated
cloud tools

Firebase (<https://firebase.google.com/>)

technology company in Asia that

Google Cloud Armor

(<https://cloud.google.com/armor/>)

Istio (<https://cloud.google.com/istio/>) big data

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

than 400 clients with business forecasts

Google Workspace

(<https://workspace.google.com/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Singapore and was the Public Sector

Google Drive

(<https://workspace.google.com/products/drive/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

More people around the world have taken to mobile gaming

(<https://www.animationxpress.com/games/75-per-cent-of-pandemic-led-mobile-gaming-surge-to-remain-static-report/>)

, including online casino games

(<https://vocal.media/gamers/the-reason-for-the-rise-in-popularity-of-online-casino-games>)

in recent years. As physical entertainment options become limited due to the pandemic, more and more people turn to their smartphones for fun and excitement. The market boom has not only attracted a growing number of gaming providers worldwide, but has increased gamers' appetite for more novel gaming experiences.

Committed to improving the online casino gaming experience, Yile Technology (<https://www.yile.com.tw>) is always optimizing the quality of its games and

introducing new game features. The Taiwanese game developer founded in 2018 has quickly become one of the country's major online casino game providers with Online808, its flagship game app that offers a wide range of popular casino games with innovative game rules. As of August 2021, the game has more than 1 million members and around 30,000 daily active users. This success has laid a solid foundation for Yile Technology to realize its ambition of entering the global market, with the team currently developing and testing a new casino game app that will be accessible for English-speaking gamers by early 2022.

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—Mezi Wu, Research and
Development Manager,
Yile Technology

When the company launched, Yile Technology deployed its games on Google Cloud (<https://cloud.google.com/>), building its infrastructure without spending too much overhead on self-owned hardware and IT personnel. At that time, the small team of 20 made an assessment that choosing a cloud platform over an on-premises environment would help save 40% of their IT operational costs and 60% of human resource costs, which would give them more resources to invest in game development and innovation. The team also knew that as Google Cloud expands its data center network worldwide, it would fall nicely alongside Yile Technology’s plans and ambition to march into the international market.

“We saw the strong potential of Google Cloud from the very beginning,” explains Mezi Wu, Research and Development Manager at Yile Technology. “By releasing top-notch software like Kubernetes and establishing an extensive, well-connected global data center network in a short time, we knew Google Cloud offered all the tools we needed to execute our global expansion plan.”

Building a cloud infrastructure that keeps delivering better performance

Yile Technology launched Online808 on Compute Engine (/compute) virtual machines (VMs), which helped handle the fluctuation of its app traffic with the automated managed instance group services. Since it was the first time that the team used cloud tools, they encountered a number of technical issues in the first month of deployment. However, the comprehensive technical support provided by Yile Technology's partner CloudMile (<https://mile.cloud/>) and the Google Cloud team ensured that the whole migration process was completed smoothly in just one month.

"We really appreciate the assistance we have received from the Google Cloud team and CloudMile," says Wu. He adds that CloudMile also even sent an engineer to Yile Technology for the entire duration of the month-long migration to troubleshoot more efficiently. "Without them, we would not have been able to launch our game in such a short time."

As Yile Technology's business grows, the CloudMile and the Google Cloud team continues to help review its cloud architecture and give advice on how to improve the performance of its services by providing thorough information about alternative cloud solutions. This saves the team a great deal of time on product surveys. As an example, Yile Technology adopted Pub/Sub (/pubsub) in 2021 to collect game records for the new games that are set to be

launched globally, replacing the third-party data pipeline tool that the company used before. The higher scalability of Pub/Sub enables the team to handle the anticipated large user growth following the release of the new games.

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*—Aries Wang, Research
and Development Deputy
Manager, Yile Technology*

Providing glitchless gaming experiences while enhancing development efficiency

Another major milestone Yile Technology took to enhance its service performance after adopting Google Cloud was to move its games from the VMs on Compute Engine to the containers on Google Kubernetes Engine (/kubernetes-engine) (GKE). When Yile Technology runs a sales promotion, the number of users on its app can increase by 3,000 in 20 minutes. With GKE, the team is able to closely monitor traffic volume on the dashboard and ensure its service stability despite sudden traffic surges through the horizontal Pod autoscaling feature.

GKE also helps the company increase its product development efficiency. When Yile Technology was still developing its games on on-premises servers, it needed to shut down the servers for 30 to 60 minutes every time it updated its apps. After moving its development workloads to GKE, the team is able to install updates with zero downtime, thanks to the

rolling update mechanism. Because GKE offers the possibility for game developers and engineers to develop apps in the same environment where they will be launched, there is no need to simulate the deployment environment at the development stage, which not only saves time but also reduces the possibility of errors that may result from moving apps to new environments.

"The automated features of Google Kubernetes Engine enables us to manage app traffic and develop games at an amazingly high level of efficiency," notes Aries Wang, Research and Development Deputy Manager at Yile Technology. "Currently, we only need two engineers to monitor traffic volume and all the environments of our three games, which frees up more workforce for development and innovation work."



Shortening game response time while ensuring quality and security

Slot games are among Yile Technology's most popular games. To provide gamers around the world with the most exciting experiences, the company needs to shorten the response time of its apps, so that the reels can stop and show the result as fast as possible after gamers hit the spin button. With its games deployed on various Google Cloud data centers around the world, Yile Technology leverages [Anthos](#) (/anthos) to manage the global network of its apps and connect gamers to the nearest server. As a result, the response time of its game has been reduced from one to two seconds to 500-600

milliseconds. Wang says that another advantage of Anthos is its simplified setup process which allows the team to build up its network management system quickly.

To further ensure the quality and security of its games, Yile Technology uses Firestore (<https://firebase.google.com/>) to record the status of its apps for troubleshooting and BigQuery ([/bigquery](https://cloud.google.com/bigquery/)) to analyze its gamers' usage records to see how certain features are received as references for improvement. It also employs Cloud Armor ([/armor](https://cloud.google.com/armor/)) to protect its games from DDoS attacks.

Facilitating remote work for more than 150 employees with Google Workspace

Yile Technology has grown its team to 150 employees since its humble beginnings and uses Google Workspace (<https://workspace.google.com/>) for internal communication and collaboration. With remote work having become the norm due to the pandemic, the company relies heavily on Google Meet (<https://workspace.google.com/products/meet/>) to showcase new features of its games in real-time, with the support of YouTube. It also leverages Drive (<https://workspace.google.com/products/drive/>) to store all the image and audio files of its games, which are mostly in large volumes, and Calendar (<https://workspace.google.com/products/calendar/>) to accommodate the work schedules of different departments.

“Google Workspace has greatly facilitated collaboration across our company by providing a one-stop work coordination solution. We don’t have to buy various software to build our own internal work system and are able to swiftly adapt to remote working, which enables our employees to focus on their work without worrying about technical issues,” says Wu.

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—Mezi Wu, Research and
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Bringing innovative games to global users with greater stability

With its two new mobile casino games scheduled for release to the global and Taiwanese market in the coming months, Yile Technology needs to ensure it can accommodate the large amount of downloads that it anticipates. The company plans to manage its global game database with Cloud Spanner (/spanner), which provides unlimited scalability across different regions, with high availability and zero scheduled downtime for maintenance.

It will also use Cloud Run (/run) to monitor the connection status of its services and game database while expanding its use of Anthos to better distribute app traffic within its global network.

Wu says, “Entering the global market has always been one of our main business goals. With the highly scalable and automated tools provided by Google Cloud, we’re confident that we can

successfully realize business expansion while continuing bringing the best possible gaming experience to gamers around the world.”

